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Branch:- AI&DS(B)

Experiment-3 : Data Ingestion and Cleaning

Aim:-DataWranglingwithPandas:Manipulateandtransform data to suit the analysis need.

Q1. What is data wrangling?. Explain the difference between Data Wrangling and Data preprocessing.

ANS:-Data wrangling istheprocessofcleaning,transforming,andorganizingrawdatainto amore usefuland structured format for analysis.It involve task such as handling missing values, filtering data, and re shaping datasets. Data preprocessing, on the other hand, is a broader trem that includes all steps taken brfore applying machine learning algorithms, including data wrangling, feature scalling, encodingcategorical data, and slpiting datasets. In simple terms, data wrangling mainly focuses on cleaning and transforming data , while data preprocessing includes additional steps required to prepare data specifically for modeling.

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', None, 'John'],
    'Age': [20, None, 22, 21],
    'Marks': [85, 90, None, 88]
})
print("Original Data:")
print(data)
data['Age'].fillna(data['Age'].mean(), inplace=True)
data['Marks'].fillna(0, inplace=True)
data.dropna(subset=['Name'], inplace=True)
print("\nAfter Data Wrangling:")
print(data)
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
data[['Marks']] = scaler.fit_transform(data[['Marks']])
print("\nAfter Data Preprocessing (Scaling):")
print(data)
```

Original Data:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	NaN	90.0
2	None	22.0	NaN
3	John	21.0	88.0

After Data Wrangling:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	21.0	90.0
3	John	21.0	88.0

/tmp/ipykernel_1395/1020328632.py:9: FutureWarning: A value is trying to be set on a copy of a DataFrame or The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=Tr

data['Age'].fillna(data['Age'].mean(), inplace=True) /tmp/ipykernel_1395/1020328632.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=Tr

```
data['Marks'].fillna(0, inplace=True)
```

After Data Preprocessing (Scaling):

	Name	Age	Marks
0	Ali	20.0	0.0
1	Sara	21.0	1.0
3	John	21.0	0.6

Q2. What is the importance of exploratory data analysis(EDA) in pandas?

ANS:- Exploratory Data Analysis (EDA) is the important step in Data analysis that helps in understanding the structure, pattern, and relationship with dataset. In Pandas, EDA allows user to summarize the data using the statistical measure, visualize trends, and detect anomalies and outliers. It helps in identifying missing values, understanding distribution, and making informed decisions about further data processing. EDA is essential because it provides insights into the data before applying any complex analysis or machine learning models.

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', 'John', 'Aman'],
    'Age': [20, 21, 22, 23],
    'Marks': [85, 90, 88, 92]
})
print(data.info())
print(data.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
 # Column   Non-Null Count  Dtype
---  -
0 Name      4 non-null      object
1 Age       4 non-null      int64
2 Marks     4 non-null      int64
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
None

count      Age      Marks
count  4.000000  4.000000
mean std  21.500000  88.750000
min
1.290994  2.986079
20.000000  85.000000
21.500000  87.500000
22.250000  90.000000
23.000000  92.000000
```

Q3. What are the main differences between series, Dataframe, and Panel?

ANS:-

Feature	Series	DataFrame
Dimensionality	1-dimensional (1D)	2-dimensional (2D)
Data Structure	Homogeneous array with a labeled index	Heterogeneous tabular data with labeled axes
Analogy	A single column of a spreadsheet	An entire spreadsheet or SQL table
Contents	Stores a single data type	Can store different data types per column
Index/Axes	One axis (index)	Two axes (index and columns)
Use Case	Storing a single feature or time series data	Most common for general tabular data
Creation	<code>pd.Series()</code>	<code>pd.DataFrame()</code>

```
import pandas as pd
# Series (1D)
s = pd.Series([10, 20, 30])
# DataFrame (2D)
df = pd.DataFrame({
    'A': [1, 2],
    'B': [3, 4]
})
print(s)
print(df)
```

```
0    10
1    20
2    30
```

```
dtype: int64
A 1B
0 3
1 2 4
```

Q4.Explain the use of:

a)isnull(): The insull() funtion is used to check the missing or null value in a dataset.It returns a boolean reslt where the turue indicates a missing value and indicates a nin-missing values

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.isnull())
```

```
      A
0  False
1   True
2  False
```

b)notnull:- The notnull() function is used to check for non-missing values in a dataset.It returns the True for values that are present and false for missing values.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.notnull())
```

```
      A
0   True
1  False
2   True
```

c)dropna():- the dropna() functon is used to removed rows or coloums that contain missing values. By default, it remove rows with at least one null value.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.dropna())
```

```
      A
0  1.0
2  3.0
```

d)fillna():- The fillns() function is used to replace the missing values with a specified values such as zero, mean, or any constant.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.fillna(0))
```

```
      A
0  1.0
1  0.0
2  3.0
```

Q5. what is column transformation and why is it important in data analysis?

ANS:- Column transformation refers to modifying or converting the values of the column to make them suitable for analysis.This may include data types, scaling values, applying mathematical operations, or creating new derived column.Column transformation is important because it improve the quality and usability of the data, making it easier to analyse and interpret.It also plays a key role in preparing data for machine learning models, where properly formatted data leads to better performance and accurate results.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.isnull())
print(df.notnull())
```

```
print(df.dropna())  
print(df.fillna(0))
```

```
      A  
0  False  
1   True  
2  False  
      A  
0   True  
1  False  
2   True  
      A  
0        
2  3..00  
      A  
0        
1  1.0  
2  0.0  
   3.0
```

Conclusion:- Data wrangling using Pandas is a crucial step in data analysis that helps convert raw data into a clean and structured format. Techniques like EDA, handling missing values, and column transformation improve data quality and make the analysis more accurate and reliable.